

AYMANE AIT DADS

Machine Learning Intern - Anomaly Detection · Time-Series Analysis · Satellite Telemetry

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PROFESSIONAL SUMMARY

Unsupervised anomaly detection specialist who built a Transformer autoencoder on Mel spectrograms achieving AUC 0.80 on industrial equipment fault detection - directly mapping to satellite telemetry monitoring requirements. At Orange Maroc, designed and deployed a Random Forest + K-Means pipeline on 100,000+ network records, reducing manual analysis time by ~60% and delivering actionable maintenance recommendations to stakeholders via Power BI dashboards. Currently pursuing a Master-level Engineering Degree in Data Science at EURECOM with coursework in Machine Learning, Deep Learning, Statistics, and Cloud Computing, with hands-on proficiency in Python, Scikit-learn, PyTorch, TensorFlow, and Git version control. Aims to apply time-series modeling, feature engineering pipeline design, and model evaluation methodology to real-world satellite telemetry datasets at ISISPACE, contributing to improved system monitoring and operational reliability.

CORE COMPETENCIES

Anomaly Detection | Time-Series Analysis | Feature Engineering | Machine Learning Models | Data Preprocessing | Model Evaluation | Data Visualization | Python & Git

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and implemented a **feature engineering pipeline** on 100,000+ fiber-optic network records - upload speed, latency, jitter - feeding automated Random Forest classification and K-Means clustering models.
- Developed an unsupervised clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for maintenance planning.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows, replacing ad-hoc analyst processes with reproducible Python scripts.
- Delivered stakeholder presentations translating model outputs and **data visualizations** (Power BI) into actionable maintenance recommendations for non-technical operations teams.

PROJECTS

Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · AUC 0.80

- Developed an **unsupervised anomaly detection** system for slide-rail machinery using a Transformer autoencoder trained on Mel spectrograms with SpecAugment, achieving AUC 0.80 with no labeled anomaly data.
- Designed the full time-series preprocessing and feature engineering pipeline - signal framing, spectrogram extraction, augmentation - then evaluated model performance and documented findings for predictive maintenance use.

PyTorch · Scikit-learn · Mel Spectrograms · SpecAugment

AI-Generated Image Detection - NTIRE

EURECOM + NTIRE 2026 International · 2025 · Ranked 1st in EURECOM Class (0.791 AUC)

2026 @ CVPR

- Ranked **1st in EURECOM class** on a private Kaggle leaderboard (0.791 AUC) and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP classification head.
- Implemented a **3-model ensemble** weighted by validation AUC with 10-view TTA on 250,000 training images; key finding: 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC), demonstrating out-of-distribution generalization.

PyTorch · OpenCLIP · FP16 Mixed Precision · AdamW

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Data Science & ML: Python | Scikit-learn | PyTorch | TensorFlow | Pandas | NumPy | Feature Engineering

Time-Series & Anomaly Detection: Transformer Autoencoders | Unsupervised Clustering | K-Means | Random Forest | Mel Spectrograms | EDA

MLOps & Visualization: Git | Linux | Power BI | Matplotlib | Seaborn | Docker (basic) | SQL